

09.09.26

$\bar{x} \in X \subseteq \mathbb{R}^n$ - ВЕКТОР СОСТОЯНИЯ

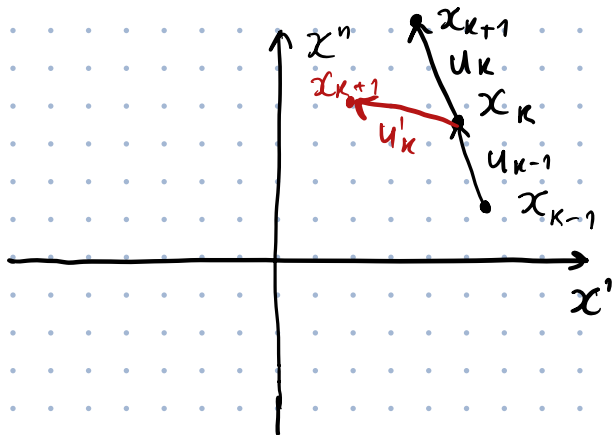
$\bar{u} \in U \subseteq \mathbb{R}^m$ - ВЕКТОР УПРАВЛЕНИЯ

$$\dot{x} = f(x, u) \quad \rightsquigarrow \quad x_{k+1} = f(x_k, u_k)$$



РАЗНОВЕ!

ВОПРОС: ПОЧЕМУ НЕ $x_{k+1} = f(x_k, x_{k-1}, u_k)$



$$x_{k+1} = f(x_k, u_k)$$

$$y_k = g(x_k) \quad // \quad y_k - \text{ВЕКТОР НАБЛЮДЕНИЯ (observation)}$$

$$\hat{x}_k = h(y_k)$$

↑ ОЦЕНКА СОСТОЯНИЯ (state estimate)

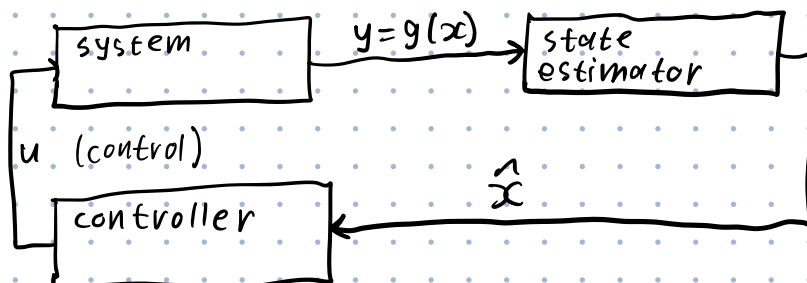
$$u_k = \mu(\hat{x}_k)$$



КОНТРОЛЛЕР

$$J = \sum_{i=0}^H G(x_{i+1}, u_i, x_{\text{target}})$$

$$(x_i^s - \tilde{x}_i^s)^2$$



КАК ДИСКРЕТИЗОВАТЬ НЕПРЕРЫВН. СИСТЕМУ?

ЛИН. СИСТ. С ПОСТ. КОЭФФ.

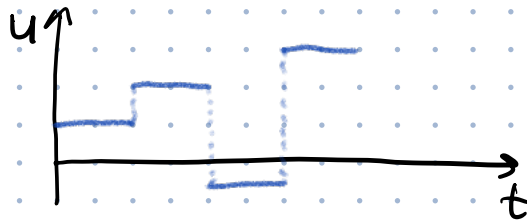
$$\dot{x} = Ax + Bu$$

sample-and-hold control

$$x(0) = x_0$$

$$u(\tau) = u_0; \tau \in [0, \Delta t)$$

$$x(\Delta t) = ?$$



ПРОЩЕ:

$$\dot{x} = Ax \quad | \quad e^{-At}.$$

$$x(0) = x_0$$

$$e^{-At} \dot{x} - e^{-At} Ax = 0$$

$$\frac{d}{dt} [e^{-At} x] = 0$$

$$e^{-At} x = C$$

$$e^{At} e^{-At} x = e^{At} C$$

$$x = e^{At} C$$

$$x(0) = x_0 = e^{A \cdot 0} \cdot C \Rightarrow C = x_0$$

$$x(t) = e^{At} x_0$$

$$x(\Delta t) = e^{A\Delta t} x_0 = (I + A\Delta t + o(\cancel{A\Delta t})) x_0 =$$

$$= x_0 + \Delta t Ax_0$$

С УПРАВЛЕНИЕМ:

$$\dot{x} = Ax + Bu \quad | \quad e^{-At}.$$

$$x(0) = x_0$$

$$u(\tau) = u_0; \tau \in [0, \Delta t)$$

$$\frac{d}{dt} [e^{-At} x] = e^{-At} Bu_0$$

$$e^{-At} x = \int_0^t e^{-A\tau} Bu_0 d\tau + C \quad | \quad e^{At}.$$

$$e^M = I + \sum_{k=1}^{\infty} \frac{M^k}{k!} = I + M + \sum_{k=2}^{\infty} \frac{M^k}{k!}$$

из лем. 4.1. $\Rightarrow C = x_0$

$$x = \int_0^t e^{A(t-\tau)} B u_0 d\tau + e^{At} x_0 =$$

$$= \int_0^t [I + A(t-\tau) + o(A(t-\tau))] B u_0 d\tau + (I + At + o(At)) x_0 =$$

$$= t I B u_0 + (I + At) x_0$$

$$x(\Delta t) = (I + A\Delta t) x_0 + \Delta t B u_0$$

Было:

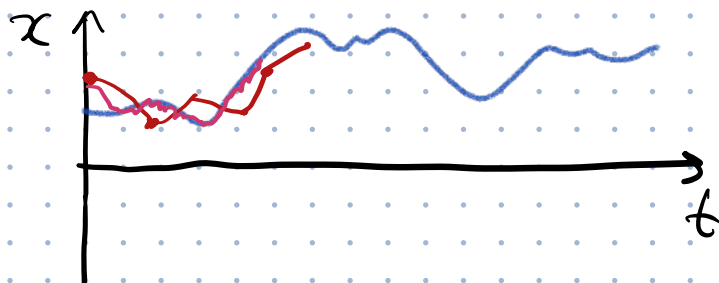
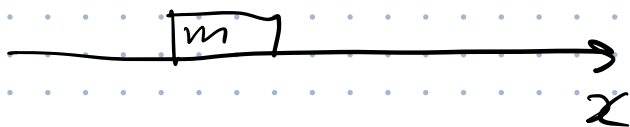
$$\dot{x} = Ax + Bu$$

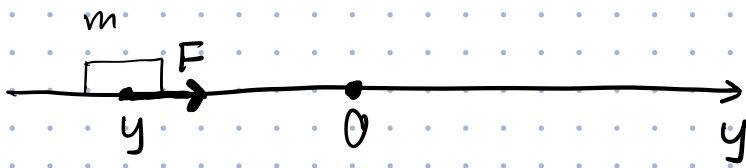
Стало:

$$x_{k+1} = \hat{A} x_k + \hat{B} u_k$$

$$\hat{A} = I + \Delta t A$$

$$\hat{B} = \Delta t B$$





$$\dot{x} = f(x, u) = Ax + Bu$$

$$\bar{x} = \begin{pmatrix} y \\ \dot{y} \end{pmatrix} \quad \ddot{y} = \frac{F}{m}$$

$$\bar{u} = (F)$$

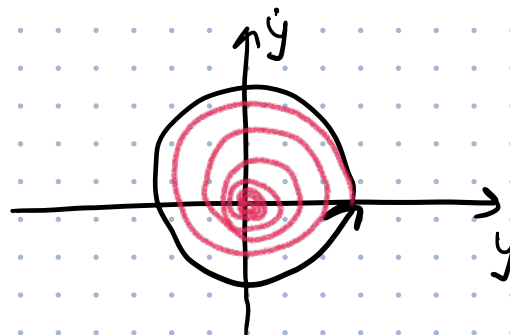
$$\boxed{\dot{\bar{x}}} = \begin{pmatrix} \dot{y} \\ \ddot{y} \end{pmatrix} = \begin{pmatrix} \dot{y} \\ \frac{F}{m} \end{pmatrix} = \begin{pmatrix} \dot{y} \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{F}{m} \end{pmatrix} =$$

$$= \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} y \\ \dot{y} \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{m} \end{pmatrix} (F) = \boxed{Ax + Bu}$$

$A \qquad x \qquad B \qquad u$

$$F = -K_p y$$

≥ 0



$$\ddot{y} = \frac{F}{m} = -\frac{K_p y}{m}$$

ДОБАВ. ДЕМПФИРОВАНИЕ

$$F = -K_p y - K_d \dot{y}$$